JUNJIE CHU



Data Analysis | SQL and Model Evaluation | Class of 2027

Available full time from late December 2026 | Based in Singapore

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Education

James Cook University Singapore | Bachelor of IT | GPA 5.85/7 **Jan 2025 - Mar 2027 expected**

Coursework: Database Modelling, Machine Learning and Data Science, Deep Learning.

EMAS Peer Tutor, May - Aug 2026: supported over 50 students in programming and IT courses.

BCA Academy Singapore | Diploma in Construction Engineering **Oct 2021 - Nov 2024**

Experience

Basilos Pte. Ltd. Singapore | AI Engineer Intern **Jul - Oct 2026**

- Updated a comparison of 20 overseas large language models using official sources; normalised versions, context windows, multimodal support, API pricing and open-source availability for technical selection.
- Organised 99 image-vectorisation experiments (36 monochrome, 60 colour, 3 hybrid); compared path counts, colour error and visual checks. Recommended simple monochrome line art and documented complex-image limits.
- Compiled 54 internal face-connector test records; labelled scenario and verification status and calculated end-to-end latency percentiles.

STAS Singapore | Operations and Administration Intern **Apr - Sep 2024**

- Maintained two-stage payment and exhibitor records for three exhibitions in Excel, categorised supplier lists, and independently researched construction-material exhibitions in China for business screening.

Selected Projects

Relational Database Design and SQL Analysis **Jun - Sep 2025**

Individual Course Projects | SQL, MySQL Workbench, ER/EER, 3NF

- Independently designed a 22-entity FlyHigh ER/EER model across fleet, booking, staff, finance and resources; defined primary/foreign keys and cardinalities and checked transitive dependencies against 3NF.
- Used JOIN, grouping and HAVING on separate library course data to identify the most-borrowed title (5236, 12 loans). For readers with at least three loans, average durations of returned loans ranged from 2 to 7 days, providing resource-planning leads.

Student Dropout Prediction and Model Comparison

Individual Course Project | Weka, EDA, K-means, DL4J MLP, 10-fold CV

- Analysed 3,630 records and 36 features; second-semester passed units distinguished dropout from graduation and formed the J48 root node. Limited findings to in-course associations, avoiding causal or pre-admission claims.
- DL4J MLP achieved weighted F1 0.893 and ROC-AUC 0.933 in 10-fold validation; detected 1,172/1,421 dropouts (recall 0.825), with 249 missed cases. Assessed false negatives alongside accuracy.

Skills

SQL (JOIN, GROUP BY, HAVING, date calculations), MySQL Workbench, Python, pandas, Excel.

EDA, feature preparation, ER/EER, 3NF, cross-validation, F1, ROC-AUC, solution cost comparison.